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RESEARCH ARTICLE

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MMP3 single-nucleotide polymorphisms are associated with noncontact ACL injuries in competing high-level athletes

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Abstract

Matrix metalloproteinases (MMPs) play an important role in matrix remodeling, as well as in ligament integrity. Anterior cruciate ligament (ACL) rupture is a severe and frequent knee injury in sports. The aim of this study was to investigate polymorphisms within the *MMP3* gene with the predisposition for noncontact ACL rupture in the Croatian professional athletes. One hundred eighty-seven (95 with ACL rupture occurring through a noncontact mechanism and 92 asymptomatic controls) unrelated Caucasians were recruited between 2016 and 2017. All participants were genotyped for three single-nucleotide polymorphisms (SNP) within the *MMP3* gene: rs591058 C/T, rs650108 A/G, and rs679620 G/A using the pyrosequencing method. For all three investigated SNPs, genotype frequencies have significantly differed between cases and controls. The MMP3 rs591058 TT ($p = 0.0012$, odds ratio [OR] = 38.541, 95% confidence interval [CI] = 1.7024–8.7254), rs650108 GG ($p = 0.0051$, OR = 23.338, 95% CI = 1.2899–4.2226) and rs679620 AA ($p = 0.0030$, OR = 34.750, 95% CI = 1.5266–7.9101) genotypes, as well as haplotype variant T-G-A ($p = 0.0104$, OR = 1.71, 95% CI = 1.13–2.59) were significantly overrepresented in cases compared to controls. These results support association between functional variants within the *MMP3* gene and the risk of ACL rupture. Still, further research is needed to corroborate these results in a larger population.

KEYWORDS

ACL Rupture, Genetic Association Study, Matrix Metalloproteinase 3, Single-nucleotide polymorphism, Sport Injury

1 | INTRODUCTION

Knee ligament injuries occur frequently during sport activities, primarily via noncontact mechanism, affecting both professional and recreational athletes. Skeletal ligaments, composed of collagenous fibers, form dense bands and are anchored to the bones at both ends, varying in size, location, and shape.[1] The anterior cruciate ligament (ACL) is one of four main knee ligaments, consisting of two major fiber bundles, anteromedial bundles (AM) and posterolateral bundles (PL). One of the most common sport injuries is the rupture of the ACL in the knee joint. In most cases, ACL injury occurs when an excessive force is applied and an excessive loading is applied to ACL.[2] Surgical reconstruction is currently most widely spread method for treatment of ACL ruptures in active patients,[3] as healing never occurs and it is the general practice for patients who develop knee instability, if not dealt in conservative manner. ACL rupture usually leads to devepoment of knee osteoarthritis.[4] ACL tears occur in 64% of athletic knee injuries and these injuries result in roughly 120,000–200,000 ACL reconstructions performed annually in United States of America, with estimated costs of 1.7 billion USD.[5] ACL

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rupture risk depends on both extrinsic and intrinsic factors. Extrinsic factors include type and intensity of physical activity, type of footwear and playing surface, medication use, nutrition, and other physiological factors. Intrinsic factors include age, gender, anatomical factors, laxity and flexibility, phase of the menstrual cycle, familial predisposition, and genetic polymorphisms. Incidence of ACL rupture is four to six times higher in women than in men[6] and individual that has suffered ACL rupture will twice more likely have ACL rupture occurrence within a family.[7] Several studies have already suggested that genetic predisposition is a significant factor in its etiology,[8] one study showed that relatives of athletes with bilateral ACL injuries had increased incidence of ACL injuries compared to unrelated controls.[9] Matrix metalloproteinases (MMPs) are members of zinc-endopeptidases that regulate cell matrix composition and can degrade extracellular matrix.[10] MMPs are composed of 25 enzymes, divided into collagenases, gelatinases, stromelysins, matrilysins or membrane-type MMPs. MMP3, member of stromelysins, is responsible for extracellular matrix remodeling and degradation, cell proliferation, and angiogenesis.[11] Single-nucleotide polymorphisms (SNPs) account for 90% of the changes and are the most common variations, which occur every 100–300 base pairs throughout the genome. Several *MMP* genes, including *MMP1*, *MMP3*, *MMP10*, and *MMP12* are located on chromosome 11 (11q22.3).[12] Several sequence variants have been associated with different conditions and disorders, such as lumbar disc degeneration,[13] osteoarthritis,[14] and rheumatoid arthritis,[15,16] as well as chronic Achilles tendinopathy[17,18] and ACL rupture.[19–21] As the different ethnic populations can exhibit different heterogeneity,[22,23] it is important to carry out research with different populations. The aim of this study was to investigate whether the polymorphisms within the *MMP3* gene are associated with the susceptibility of ACL ruptures in Croatian subpopulation.

2 | METHODS

Our study is case–control study (Level 3).

2.1 | Participants

In this case–control genetic association study a total of 187 participants were recruited between 2016 and 2017. They were all either recreationally competing or professional athletes, self-reported Caucasians and unrelated. The ACL rupture group consisted of 95 participants (69 male and 26 female, average age 25.8 years), diagnosed with magnetic resonance and arthroscopically confirmed ACL rupture through the noncontact mechanism of the injury. All ACL participants were active competing athletes (soccer, handball, basketball, martial arts, skiing), having between three and seven coach supervised trainings per week, recruited at the time of the injury in the Orthopedic Clinic or by a team doctor specialized in orthopedic surgery. Control group consisted of 92 participants (72 male and 20 female, average age 39.0 years) that have finished their active athlete career and have reported not sustaining any ligament ruptures throughout their athlete career. Main selection criteria for the control group were recruited athletes who retired and have finished their professional career without sustaining ACL injury. Through their professional career control group was closely monitored by obligatory medical exams every year and their club doctors if they had any injury. In the time of the recruitment, they were all without any self-reported injuries. If someone from control group had any symptoms of knee injury at the time of recruitment orthopedic surgeon with over 30 years of clinical experience with sports and knee injury would do clinical exam and tests. Participants from control group were not subjected to MRI exam or knee arthroscopy for the purpose of this study. Control recruitment was monitored by two experienced observers to ensure background quality of the recruited controls. Orthopedic surgeon has over 30 years of clinical experience with sports and knee injury, working as chief officer at football club and national football team, his area of expertise being knee ligament reconstruction and retired professional athlete who is working as a professor at the Faculty of Kinesiology, specialized in biomechanics of movement and injury prevention exercise programs. This study was approved by the medical Ethics committee of the School of Medicine and by the Ethics committee of the Faculty of Kinesiology, University of Zagreb where the research was done. All participants have given written informed consent and have provided a biological sample for analysis.

2.2 | Genotyping

Genomic DNA was extracted from the oral epithelial cells using QIAamp DNA kit (Qiagen), followed by PCR amplification using MMP3 rs591058, rs650108, and rs679620 specific primer pairs (Metabion) and PyroMark PCR Kit (Qiagen). Amplified fragments were sequenced by pyrosequencing method (PyroMark Q24; Qiagen) according to the manufacturer's protocol and analyzed by Pyromark Q24 software.

2.3 | Statistical analysis

An odds ratio method was used for allelic, genotypic, and haplotype differences, respectively, using Statcalc program, AcaStat software. A statistically significant difference was defined when p was $p < 0.05$. Haplotype analysis was performed using Phase software. The Hardy–Weinberg equilibrium and linkage disequilibrium (LD) analyses were done by the Arlequin software version 3.5 (Genetics and Biometry Laboratory, University of Geneva, Geneva, Switzerland). In this study, we used groups similar in size to previously reported studies[20,22,24,25] investigating genotype effects on various soft-tissue injuries, as those group sizes proved to be large enough to detect significant results, so power analysis was not performed. The Bonferroni correction was not used in this study, as it is considered

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too conservative.[26] *P* values were adjusted for false discovery rate (FDR) using Benjamini–Hochberg procedure for adjusted *p* value. It was applied for each genotypic, allelic, and haplotype separately.[27]

3 | RESULTS

3.1 | Participants characteristics

Participants were divided into two groups: self-reported noncontact mechanism of ACL rupture occurrence and control group that has not self-reportedly suffered from any ligament injuries during their active years in competing professional sport. The control group was significantly older and heavier, with the higher BMI index than ACL group, but that is explained by the different lifestyle of the active professional athletes and retired athletes. Due to the similar average heights, we have assumed weight and BMI were similar between ACL rupture group and controls, when controls were active competing athletes, but at the presented data are not matched for weight and BMI, only for height, ethnicity, and gender (Table 1).

Genotype frequencies were significantly different between groups, TT genotype of rs591058, GG genotype of rs650108, and AA genotype of rs679620 were significantly overrepresented in ACL rupture group and associated with predisposition for ACL injury. Similarly, allelic frequencies significantly differed between groups. C allele of rs591058, A allele of rs650108 and G allele of rs679620 were significantly overrepresented in controls compared to cases and therefore classified as protective, while T allele of rs591058, G allele of rs650108 and A allele of rs679620 were significantly overrepresented in cases comparing to controls thus being considered as a predisposition for injury (Table 2).

All polymorphisms conformed to the Hardy–Weinberg equilibrium in both the ACL rupture and the control group (Table 2). SNPs rs591058, rs650108, and rs679620 were also in the LD, $D' = 1.000$; $r^2 = 0.4014$ and $D' = 0.8893$; $r^2 = 0.3242$, respectively. Apart from Bonferroni correction being considered too conservative for these types of studies, there is no need for it, as all three SNPs are in LD, meaning they are inherited as blocks.

Haplotypes constructed from all possible combinations of polymorphisms were analyzed (only relevant data is shown in Table 3). As all three polymorphisms are in LD, it was expected for T-G-A haplotype to be significantly overrepresented in cases, being associated with predisposition for ACL rupture, while C-A-G haplotype was significantly overrepresented in controls, being associated with protection.

TABLE 1 General characteristics of the ACL rupture (ACLR) group and control group.

	ACLR (*n* = 95)	Controls (*n* = 92)	*p* Value
Age (years)	25.8 ± 8.3	39.0 ± 11.4	**0.0001**
Height (cm)	178.8 ± 7.3	179.8 ± 9.6	0.4349
Weight (kg)	77.5 ± 11.8	84.6 ± 15.4	**0.0005**
BMI (kg/m^2)	24.2 ± 2.9	26.0 ± 3.3	**0.0001**
Ethnicity (Caucasian)	100% (95)	100% (92)	1.0000
Gender (% male)	73% (95)	78% (92)	0.4270

Note: Age, height, and weight are self-reported values in time of the recruitment. Ethnicity and gender are represented as a percentage, the remaining variables are expressed as a mean ± standard deviation. Significant *p* values are in bold.

4 | DISCUSSION

The main finding of this study was the allelic and genotype association of the investigated SNPs, TT genotype of rs591058, GG genotype of rs650108, and AA genotype of rs679620 within *MMP3* gene for the ACL rupture, as well as their haplotypes, showing a significant association with the risk of ACL rupture in Croatian subpopulation.

The *MMP* gene family codes for over 20 different zinc-dependent endopeptidases. MMP3 is a proteoglycan-degrading enzyme that plays a significant role in degradation of collagens and other structural components, mainly proteoglycans, fibronectin, laminin, aggrecan, and gelatine.[28] Out of 23 MMPs in humans, 9 of them form a cluster on the long arm of chromosome 11, resulting in an almost perfect LD of rs591058 and rs679620, and both being in high LD with the third rs650108.[18]

The MMP3 SNP rs679620 was previously associated with ACL rupture as part of the inferred haplotype in a South African cohort, together with the with the MMP10 rs486055C/T, MMP1 rs1799750 1G/2G, and MMP12 rs2276109 A/G polymorphisms, but there was no independent association for MMP3 rs679620, though GG genotype had a trend toward overrepresentations in controls,[29] which is in accordance with the data we have presented in our study, having statistically significant overrepresentation of GG genotype in our controls. In an additional study of South African cohort, investigating same MMP3 SNPs included in the present study, there were no significant associations between MMP3 SNPs and ACL group where rupture originated from both contact and noncontact mechanism of injuries,[22] possibly due to a nonhomogeneous group.

The 5A/6A polymorphism in the promoter region of the *MMP3* gene affects the regulation of *MMP3* gene expression. Study of the Thailand cohort found a significant difference for rs3025058 (5A/6A, MMP3 promoter) in genotype 5A/5A between the ACL patients with contact sport and without contact sport, while in allelic association significant differences were 6A: 81.2% in contact sports versus 89.1% in noncontact sports and 5A: 18.8% in contact sports versus 10.9% in noncontact sports.[20] More recent study of the Polish cohort found the MMP3 rs679620 G allele and the MMP3 rs591058 C allele could predispose for ACL ruptures in physically active Caucasians,[19] which is contradictory to our findings.

Different risk associations have already been reported in populations with independent ancestry, and even though the reason

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TABLE 2 Allele and genotype frequency for MMP3 SNP.

MMP3			Allele frequency					
SNP rs591058	**C>T**	**ACLR (*n* = 95)**	**CON (*n* = 92)**	***p* Value**	**FDR**	**OR**	**95% CI**	**Association**
1	C	48.9% (91)	62.5% (115)	**0.0047**		0.5515	0.3651–0.8331	**Protection**
2	T	52.1% (99)	37.5% (69)	**0.0047**	**0.0047**	1.8132	1.2004–2.7388	**Predisposition**
			Genotype frequency					
SNP rs591058	**C>T**	**ACLR (*n* = 95)**	**CON (*n* = 92)**	***p* Value**	**FDR**	**OR**	**95% CI**	**Association**
11	CC	25.3% (24)	34.8% (32)	0.1567		0.6338	0.3372–1.1913	None
12	CT	45.3% (43)	55.4% (51)	0.165		0.6648	0.3736–1.1831	None
22	TT	29.4% (28)	9.8% (9)	**0.0012**	**0.0024**	38.541	1.7024–8.7254	**Predisposition**
HWE		0.364	0.080					
MMP3			**Allele frequency**					
SNP rs650108	**A>G**	**ACLR (*n* = 95)**	**CON (*n* = 92)**	***p* Value**	**FDR**	**OR**	**95% CI**	**Association**
1	A	27.4% (52)	39.1% (72)	**0.0161**		0.5862	0.3793–0.9057	**Protection**
2	G	72.6% (138)	60.9% (112)	**0.0161**	**0.0153**	1.706	1.1041–2.6362	**Predisposition**
			Genotype frequency					
SNP rs650108	**A>G**	**ACLR (*n* = 95)**	**CON (*n* = 92)**	***p* Value**	**FDR**	**OR**	**95% CI**	**Association**
11	AA	9.5% (9)	12% (11)	0.5836		0.7706	0.3035–1.9566	None
12	AG	35.8% (34)	54.3% (50)	**0.0112**	**0.0168**	0.4682	0.2604–0.8419	**Protection**
22	GG	54.7% (52)	33.7% (31)	**0.0051**	**0.0161**	23.338	1.2899–4.2226	**Predisposition**
HWE		0.331	0.177					
MMP3			**Allele frequency**					
SNP rs679620	**G>A**	**ACLR (*n* = 95)**	**CON (*n* = 92)**	***p* Value**	**FDR**	**OR**	**95% CI**	**Association**
1	G	48.4% (92)	62.5% (115)	**0.0064**		0.5633	0.3729–0.8507	**protection**
2	A	51.6% (98)	37.5% (69)	**0.0064**	**0.009**	1.7754	1.1754–2.6815	**predisposition**
			Genotype frequency					
SNP rs679620	**G>A**	**ACLR (*n* = 95)**	**CON (*n* = 92)**	***p* Value**	**FDR**	**OR**	**95% CI**	**Association**
11	GG	24.2% (23)	34.8% (32)	0.1142		0.5990	0.3171–1.1313	None
12	GA	48.4% (46)	55.4% (51)	0.3376		0.7547	0.4246–1.3416	None
22	AA	27.4% (26)	9.8% (9)	**0.0030**	**0.0064**	34.750	1.5266–7.9101	**Predisposition**
HWE		0.765	0.080					

Note: Allele and genotype frequencies are expressed as percentage with the number of participants (*n*) in parentheses. Results that have been classified as significant by the *p* value ($p < 0.05$) in bold and have been additionally checked for false discovery rate (FDR), using Benjamini–Hochberg adjusted p value.

Abbreviations: CI, confidence interval; HWE, Hardy–Weinberg equilibrium; OR, odds ratio.

TABLE 3 Haplotype frequency.

Hap code	Haplotype	CON	ACLR	*p* Value	FDR	OR	95% CI	Association
1	C-G-G	43	40	0.51		0.85	0.52–1.39	None
2	C-A-G	71	50	**0.0074**	**0.0222**	0.55	0.35–0.85	**Protection**
3	C-A-A	0	3	–	–	–		
4	T-G-G	0	2	–	–	–		
5	T-G-A	68	97	**0.0104**	**0.0156**	1.71	1.13–2.59	**Predisposition**

Note: Haplotype Hap code 3 and 4 were not predicted by PHASE in our samples. Results that have been classified as significant by the *p* value ($p < 0.05$) in bold and have been additionally checked for false discovery rate (FDR), using Benjamini–Hochberg adjusted *p* value.

Abbreviations: CI, confidence interval; OR, odds ratio.

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is not fully understood, it has been hypothesized that study participants indirectly implicate a region where biologically informative variants can be found.[23] Apart from independent ancestry, the main difference in our study is that our control group that was not matched for age and weight with the ACL rupture group. We wanted to make sure controls never suffered from ligament injuries through their active competing career in professional sport, to have a real control group. If control group were active competing athletes, we could not be sure they would not suffer from ACL injury in their active sports future. To eliminate that, we decided to recruit participants after their competing sport career ended, so they are older than cases and their body weight is somewhat higher, but that does not reflect their body composition when they were actively competing. It is fair to assume their weight and BMI would be matched to an active ACL rupture group.

The Polish cohort had a group of recreationally active controls that cannot match volume of training when compared to professional athletes, and controls were matched for age with the ACL rupture group, aging only ~25 years, so it cannot be known whether they will suffer from ACL rupture in the future, especially as they do not have same training load as the ACL rupture group.

Taking into the account the importance of investigating a noncontact mechanism of injury as they occur due to the athlete's own movements, without any external force acting, we have included only noncontact mechanism of injury in our ACL rupture group. In a noncontact injury, that has higher intrinsic component than a contact injury, mechanisms are classified as sudden deceleration before the change of direction or before landing after jump. Contact injuries involve contact with another person, usually via collision or accident or unknown source.[30] Cost of ACL rupture is from $17,000 to $25,000 per injury, with conservative estimates of surgery and rehabilitation.[31] ACL injuries are particularly significant as 80% of all knee ligament injuries result in surgery and in general population sport activities account for 65% of those injuries.[32] Additional loss for professional athletes is absence from trainings, games, and competitions. Sport clubs also suffer significant financial losses. In some cases, injured athlete needs to be temporarily replaced with uninjured one, and that causes additional financial issues in professional sport. The results presented in this study could be used in future clinical practice when interpreted with caution. Limitations of this study are the small sample size, so any interpretation of the results should be done with caution. Gender distribution between groups was matched, but both consisted of a larger proportion of male cases, even though it was previously reported that females are in greater risk.[33–35] To our knowledge, this is a first study that has investigated MMP3 SNPs related to a noncontact injury of ACL in competing athletes, while having a control group of retired competing athletes who have finished their sport careers without ACL injuries. Future studies should recruit larger number of participants in both groups but should also consider the significance of controls that have had all the possibilities to suffer an injury and they have still finished their active career without them. Though they are not matched for age and weight with injury group, their genetic makeup gives a better picture than control groups consisting of active recreational athletes whose future and injury prospective is unknown in the long run.

5 | PERSPECTIVE

Genetic studies are novel field in the sports medicine. Our research is small step forward in understanding intrinsic risk factors for ACL rupture. Many books and articles wrote about connection of genes and ACL ruptures, but scientific evidence was poor.[7–9] This study showed genetic association between genotypes (TT genotype of rs591058, GG genotype of rs650108, and AA genotype of rs679620) and ACL rupture in professional athletes. With this knowledge we can develop "easy to use" screening tests for ACL rupture (ligament injury) in young athletes who have increased risk for injury. In this way coach or club can introduce preventive exercises, correct wrong biomechanical patterns and introduce better recovery protocols for young competing athletes with positive screening test for ACL rupture.

6 | CONCLUSION

There is an increased amount of genetic case–control association studies trying to identify genetic sequence variants within genes coding for soft tissue building blocks and extracellular matrix to associate them with the risk of developing soft tissue injuries. This study investigated one small fraction of variants within the gene coding for proteoglycan-degrading enzyme and whether three functional variants of the MMP3 gene are associated with a noncontact manner of ACL rupture in competing athletes. A novel finding of this study are significant associations of investigated genotypes (TT genotype of rs591058, GG genotype of rs650108, and AA genotype of rs679620) that were significantly overrepresented in ACLR group and associated with predisposition for ACL injury.

AUTHOR CONTRIBUTIONS

Nina Simunic-Briski, Goran Vrgoc, Zlatko Dembic, and Gordan Lauc substantially contributed to research design, acquisition, analysis, and interpretation of data, drafting the paper and revising it critically, approval of the submitted and final versions. All authors have read and approved the final submitted manuscript. Sasa Jankovic and Damir Knjaz substantially contributed to acquisition, analysis, and interpretation of data, drafting the paper and revising it critically, approval of the submitted and final versions.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

ETHICS STATEMENT

The study was approved by the Medical ethics committee of the School of Medicine, University of Zagreb and by the Ethics committee of the Faculty of Kinesiology, University of Zagreb.

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